Example 6.31

A manufacturer has to supply 12,000 units of a product per year to his customer. The ordering cost (C_3) is ₹ 100 per order and carrying cost is ₹ 0.80 per item per month. Assuming there is no shortage cost and the replacement is instantaneous, determine the

(i) economic order quantity
(ii) time between orders
(iii) number of orders per year

Solution:

Demand per year: R = 12,000 units

Ordering cost: C_3 = ₹ 100/order

Carrying cost: C_1 = 0.80/item/month
= 0.80×12 per year
= ₹ 9.6 per year

(i) $EOQ = \sqrt{\frac{2C_3R}{C_1}} = \sqrt{\frac{2\times100\times12000}{9.6}}$
= 500 units

(ii) Number of order per year = $\frac{Demand}{EOQ}$
= $\frac{12,000}{500}$ =24

(iii) Optimal time per order = $\frac{1}{t_0} = \frac{1}{24}$ year
= $\frac{12}{24}$ months = $\frac{1}{2}$ month = 15 days

Example 6.32

A company has to supply 1000 item per month at a uniform rate and for each time, a production run is started with the cost of ₹ 200. Cost of holding is ₹ 20 per item per month. The number of items to be produced per run has to be ascertained. Determine the total of setup cost and inventory carrying cost if the run size is 500, 600, 700, 800. Find the optimal production run size using EOQ formula.

Solution:

Demand : R = 1000 per month
Setup cost : C_3 = ₹ 200 per order
Carrying cost: C_1 = ₹ 20 per item per month.

Run size q	Set up cost $\frac{R}{q}\times C_3$	Carrying cost $\frac{q}{2}\times C_1$	Total Inventory cost (Set up cost + Carrying cost)
500	$\frac{1000}{500}\times200 = 400$	$\frac{500}{2}\times20 = 5000$	5400
600	$\frac{1000}{600}\times200 = 333.3$	$\frac{600}{2}\times20 = 6000$	6333.3
700	$\frac{1000}{700}\times200 = 285.7$	$\frac{700}{2}\times20 = 7000$	7285.7
800	$\frac{1000}{800}\times200 = 250$	$\frac{800}{2}\times20 = 8000$	8250

Table : 6.3

$EOQ = \sqrt{\frac{2RC_3}{C_1}} = \sqrt{\frac{2\times1000\times200}{20}}$

$= \sqrt{20000}$ = 141 units (app.)

Example 6.33

A manufacturing company has a contract to supply 4000 units of an item per year at uniform rate. The storage cost per unit per year amounts to ₹ 50 and the set up cost per production run is ₹ 160. If the production run can be started instantaneously and shortages are not permitted, determine the number of units which should be produced per run to minimize the total inventory cost.

Solution :

Annual demand: R = 4000

Storage cost: C_1 = ₹ 50

Setup cost per production: C_3 = ₹160

$EOQ = \sqrt{\frac{2Rc_3}{c_1}} = \sqrt{\frac{2\times4000\times160}{50}} = 160.$

$\therefore$ To minimize the production cost, number of units produced per run is 160 units.

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